

ReplicationPad3d#

[https://pytorch.org/docs/stable/generated/torch.nn.ReplicationPad3d.html#tor](https://pytorch.org/docs/stable/generated/torch.nn.ReplicationPad3d.html#torch.nn.ReplicationPad3d)

Description:

Pads the input tensor using replication of the input boundary. For N-dimensional padding, use `torch.nn.functional.pad()`. padding (int, tuple) – the size of the padding. If is int, uses the same padding in all boundaries. If a 6-tuple, uses (`padding_left\text{padding\_left}`padding_left, `padding_right\text{padding\_right}`padding_right, `padding_top\text{padding\_top}`padding_top, `padding_bottom\text{padding\_bottom}`padding_bottom, `padding_front\text{padding\_front}`padding_front, `padding_back\text{padding\_back}`padding_back) Note that the output dimensions must remain positive. Input: (N,C,D_{in},H_{in},W_{in})(N, C, D_{in}, H_{in}, W_{in})(N,C,D_{in},H_{in},W_{in}) or (C,D_{in},H_{in},W_{in})(C, D_{in}, H_{in}, W_{in})(C,D_{in},H_{in},W_{in}). Output: (N,C,D_{out},H_{out},W_{out})(N, C, D_{out}, H_{out}, W_{out})(N,C,D_{out},H_{out},W_{out}) or (C,D_{out},H_{out},W_{out})(C, D_{out}, H_{out}, W_{out})(C,D_{out},H_{out},W_{out}), where

Function Signature

```
class torch.nn.ReplicationPad3d(padding)[source]#
```

Parameters

Shape:

Code Examples

```
>>> m = nn.ReplicationPad3d(3)
>>> input = torch.randn(16, 3, 8, 320, 480)
>>> output = m(input)
>>> # using different paddings for different sides
>>> m = nn.ReplicationPad3d((3, 3, 6, 6, 1, 1))
>>> output = m(input)
```

Detailed Documentation

Documentation

Pads the input tensor using replication of the input boundary.

For N-dimensional padding, use `torch.nn.functional.pad()`.

padding (int, tuple) – the size of the padding. If is int, uses the same padding in all

boundaries. If a 6-tuple, uses $(padding_left, padding_right, padding_top, padding_bottom, padding_front, padding_back)$. Note that the output dimensions must remain positive.

Input: $(N, C, D_{in}, H_{in}, W_{in})$ or $(C, D_{in}, H_{in}, W_{in})$.

Output: $(N, C, D_{out}, H_{out}, W_{out})$ or $(C, D_{out}, H_{out}, W_{out})$, where

$D_{out} = D_{in} + padding_front + padding_back$

$H_{out} = H_{in} + padding_top + padding_bottom$

$W_{out} = W_{in} + padding_left + padding_right$